American Computer Science League

2024-2025 ● Contest 3: Short Problems Solutions ● Senior Division

1. Boolean Algebra

$A\,(\overline{B} + \overline{C}) + \overline{A\,\overline{B}\,C} + \overline{B\,C}$
$= A\overline{B} + A\overline{C} + \overline{A} + B + \overline{C} + \overline{B} + \overline{C}$
$= \overline{B}(A + 1) + \overline{C}(A + 1 + 1) + \overline{A} + B$
$= \overline{B} + \overline{C} + \overline{A} + B$
$= (\overline{B} + B) + \overline{C} + \overline{A}$
$= 1$

This expression is always TRUE, so 8 triples make it TRUE.

E. 8

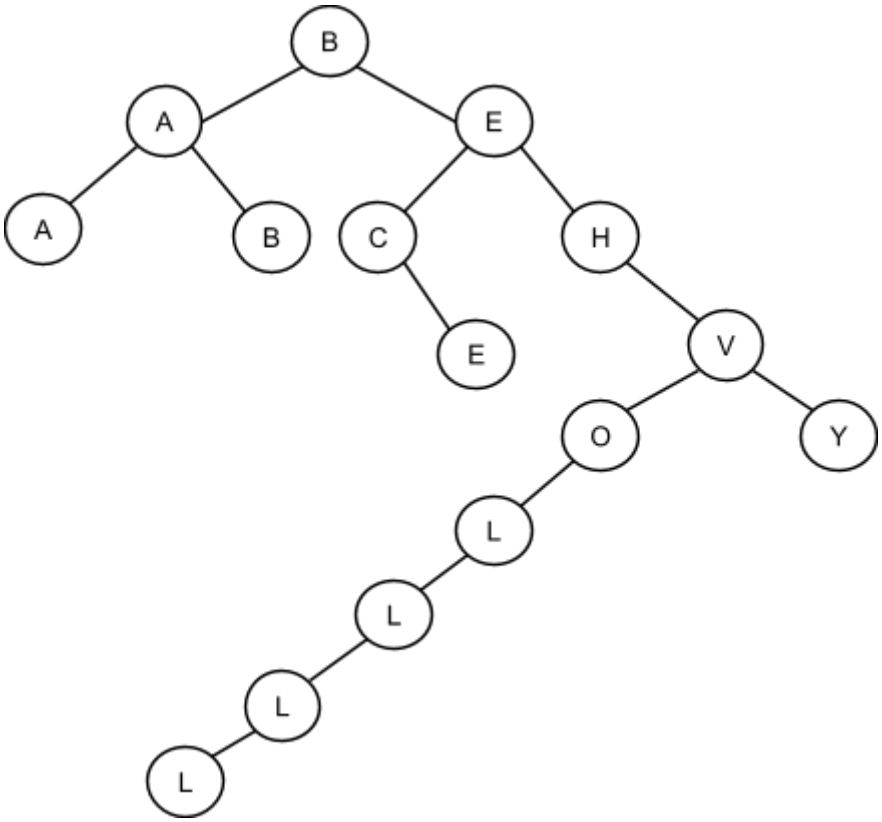
2. Boolean Algebra

$\overline{A(B + \overline{C})}\,(A\,\overline{B\,C})\,(\overline{A} + \overline{B})$
$= (\overline{A} + \overline{B + \overline{C}})(A(\overline{B} + \overline{C}))(\overline{A} + \overline{B})$
$= (\overline{A} + \overline{B}C)(A\overline{B} + A\overline{C})(\overline{A} + \overline{B})$
$= (\overline{A}A\overline{B} + \overline{A}A\overline{C} + \overline{B}CA\overline{B} + \overline{B}CA\overline{C})(\overline{A} + \overline{B})$
$= A\overline{B}C(\overline{A} + \overline{B})$
$= A\overline{B}C\overline{A} + A\overline{B}C\overline{B}$
$= A\overline{B}C$

C. $A * \overline{B} * C$

3. Data Structures

The binary search tree for **BEACHVOLLEYBALL** is:



There are 6 nodes with only 1 child: C, H, O, L, L, L

C. 6

4. Data Structures

Starting at the root node: M

- ? is to the right of M, so ? > M
- ? is to the left of T, so M < ? ≤ T
- ? is to left of Q, so M < ? ≤ Q

Therefore the choice is A. M < ? ≤ Q

A. M < ? ≤ Q

5. FSAs

a. abbada satisfies the upper path
b. abaccabb fails on the upper path at the second a
c. acddcab satisfies the lower path
d. acca satisfies the lower path

D. a, c, d

6. Regular Expressions

[a-cg-p]* [^s]* [^o-t] [d-ho-y]*

checkers	chess	monopoly	cribbage
bingo	uno	racko	duo

All 8 satisfy the regular expression.

A. all of them
B. 7
C. half of them
D. 1
E. none of them